LECTURE NOTE 5: Advance VLAN Configuration

1. VLAN Membership: Static vs. Dynamic

VLAN membership defines how devices are assigned to specific VLANs within a network. There are two primary methods for VLAN membership: static and dynamic.

a. Static VLAN Membership (Port-based VLANs)

In a static VLAN, a device is manually assigned to a specific VLAN by configuring the switch port to which the device is connected. This method is also known as port-based VLAN configuration.

The network administrator manually assigns a port to a specific VLAN. All traffic from the device connected to that port is tagged with the corresponding VLAN ID.

Example, if Port 1 is assigned to VLAN 10, any device connected to Port 1 will belong to VLAN 10.

Advantage

- ✓ Simple and easy to configure for small networks.
- ✓ Provides high control over which devices are assigned to specific VLANs.

Disadvantage

- ✓ Not scalable for large networks with frequent device changes.
- ✓ Requires manual configuration if devices move between different network ports.

Example of Configuration

```
Switch(config)# interface fastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

b. Dynamic VLAN Membership

In a dynamic VLAN, the VLAN membership of a device is automatically determined based on the device's MAC address.

Dynamic VLAN Assignment relies on a VLAN Management Policy Server (VMPS). When a device connects to the switch, the switch queries the VMPS server to determine which VLAN the device should be assigned to, based on the device's MAC address.

The device is automatically placed into the correct VLAN, and no manual configuration on individual switch ports is needed.

Advantages

- ✓ Ideal for environments where devices frequently move between ports (e.g., employees moving to different desks).
- ✓ Reduces the administrative burden of manually configuring VLAN assignments.

Disadvantages

- ✓ More complex to set up compared to static VLANs.
- ✓ Requires a VMPS server, which adds to the cost and complexity of the network.

Dynamic VLAN Process

- ✓ The device connects to the network.
- ✓ The switch queries the VMPS server with the device's MAC address.
- ✓ The VMPS server responds with the appropriate VLAN assignment.
- ✓ The switch dynamically assigns the port to the correct VLAN.

2. VLAN Management

VLAN management involves configuring, monitoring, and maintaining VLANs within a network. Proper VLAN management ensures that the network is segmented effectively and operates securely and efficiently.

a. VLAN Naming and Documentation

VLAN Naming:

- ✓ Each VLAN should have a clear, descriptive name that reflects its purpose (e.g., HR_VLAN, SALES_VLAN).
- ✓ This improves network visibility and helps administrators easily identify the VLANs in use.

Documentation:

- ✓ Documenting the VLAN configuration is critical for future network management. It includes the VLAN IDs, names, ports associated with each VLAN, and any dynamic VLAN policies.
- ✓ Having well-maintained documentation prevents errors and simplifies troubleshooting.

b. VLAN Trunking and Management Across Switches

Trunk Links:

- ✓ Trunk links allow traffic from multiple VLANs to pass between switches. A trunk port is configured to carry traffic for all or specific VLANs using 802.1Q VLAN tagging.
- ✓ A single physical link can carry multiple VLANs, which is essential in larger networks with many interconnected switches.

Management of VLANs Across Multiple Switches:

- ✓ VLAN configurations must be consistent across switches to maintain proper network segmentation.
- ✓ VLAN information is propagated using protocols like VTP (VLAN Trunking Protocol), which helps manage VLAN configuration across the entire network.

c. VLAN Trunking Protocol (VTP)

- ✓ VTP is used to propagate VLAN configurations automatically across multiple switches within the same VTP domain. It simplifies VLAN management in larger networks.

VTP Modes:

- ✓ ***Server Mode***: The switch can create, modify, and delete VLANs. Changes made on a VTP server are propagated to all other switches in the VTP domain.
- ✓ ***Client Mode***: The switch receives VLAN information from VTP servers but cannot make changes.
- ✓ ***Transparent Mode***: The switch does not participate in the VTP domain but forwards VTP advertisements between other switches.

VTP Configuration :

```
Switch(config)# vtp mode server
Switch(config)# vtp domain NetworkDomain
Switch(config)# vtp password securepassword
```

d. VLAN Security and Best Practices

VLAN Hopping Attack Prevention:

- ✓ VLAN hopping is a security exploit where an attacker gains access to traffic in different VLANs.
- ✓ Best practices include disabling unused switch ports and ensuring trunk ports are properly configured (limiting allowed VLANs).

Best Practices:

- ✓ Disable unused ports: Shut down unused switch ports to prevent unauthorized access.
- ✓ Limit VLANs on trunk ports: Only allow necessary VLANs on trunk links to minimize security risks.
- ✓ Use Native VLAN carefully: The native VLAN should be used cautiously as it transmits untagged frames. Consider changing it from the default VLAN 1 to a less frequently used VLAN for security purposes.